

Radar model card, resolution budget, and public-proxy boundary

Branch	Band	BW MHz	dR m	PRF/CRF Hz	CPI ms	Pulses	dfD Hz	dv m/s
High-res. X/Ku	X/Ku	600	0.250	4000/4000	6.0	24	166.7	2.498
Tactical S-band	S	180	0.833	4000/4000	6.0	24	166.7	8.059
GBAD 3D/4D	X/Ku cue	300	0.500	4000/4000	6.0	24	166.7	2.426

Public-proxy assumptions

positive class: fixed-wing pusher-prop
public proxy
geometry: swept/delta wing; rear pusher propulsor
speed: 45--60 m/s; prop micro-Doppler: 150--217 Hz
phase windows: 0--30 s, 30--90 s, 90--150 s
launch proxy: rail/catapult take-up
RCS/aspect: -28 to -2 dBsm public-source proxy

Receiver impairments

AGC	PRF ambiguity
compression	
clock drift	Doppler folding
quantization	calibration offset
dropped CPI	multipath masking

Cue definitions

acoustic: node-wise spectral cadence, cross-node agreement, and amplitude stability
passive RF: no-signal / RFI burst / clock-offset / provenance-quality geometry
claim boundary: public-proxy assumptions only

All values are public-proxy or synthetic archetype assumptions; no measured-platform truth is implied.